

Comprehending ER diagrams and UML class diagrams

TASK: T2

TREATEMENT: CD_EX

First Name:.....

Last Name:.....

ID:.....

1. Comprehending

Start Time:

Figure 1 shows the UML class diagram of a system managing a company. In particular, the diagram regards the company resources and projects information.

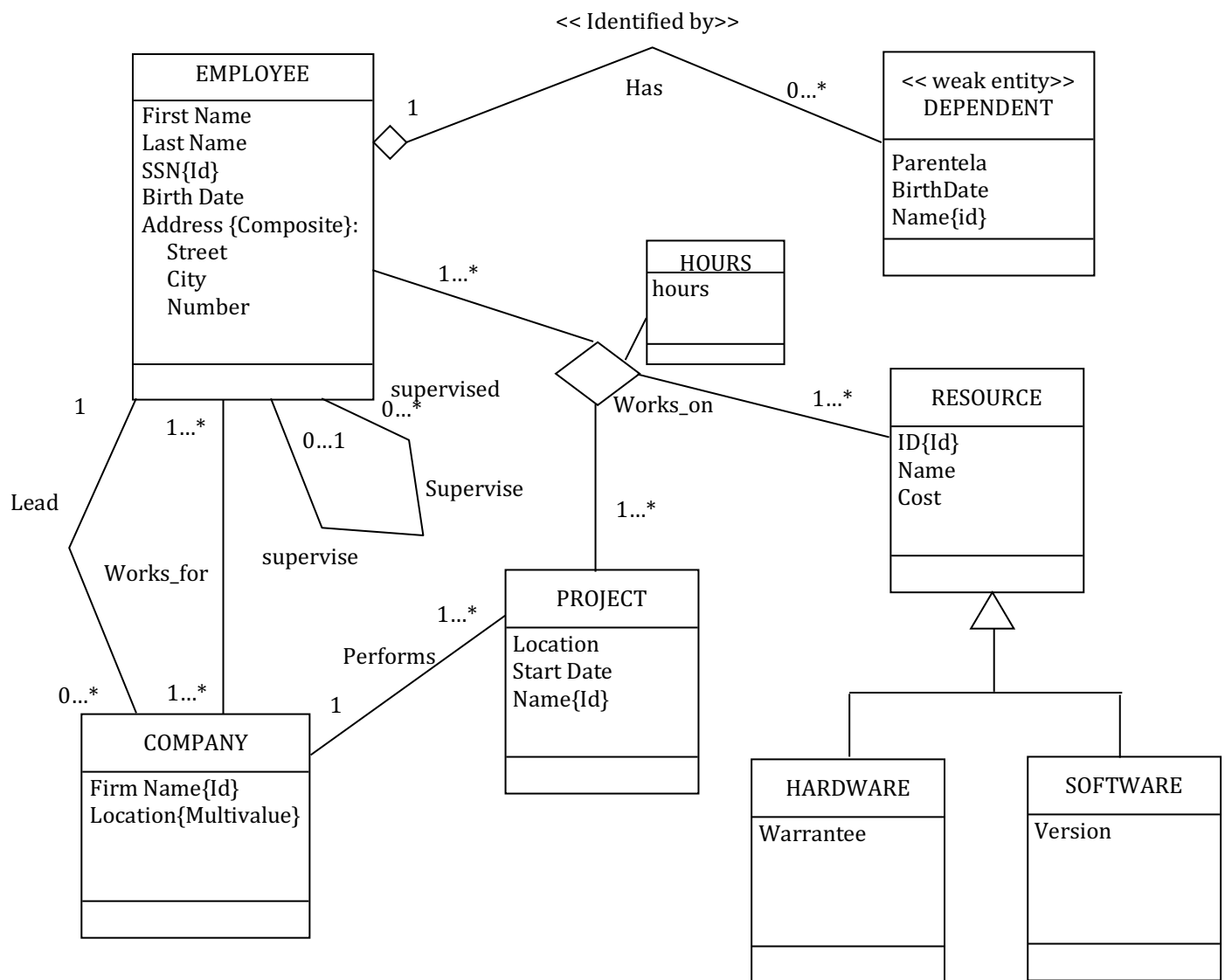


Figure 1: The UML class diagram of a system managing company information

Analyze the diagram in Figure 1 and answer the following questions.

Q1 Focusing on class Employee, which of the following entries is true?

- ☐ Given an employee, it is possible to obtain his/her hiring date.
- ☐ Given an employee, it is possible to obtain his/her birth date
- ☐ Given an employee, it is possible to obtain his/her salary

Q2 Focusing on class Project, which of the following entries is true?

- ☐ A project is univocally identified by its name
- ☐ A project is univocally identified by its and location
- ☐ A project is univocally identified by its name and start date

Q3 Focusing on class Employee, which of the following entries is true?

- ☐ Given an employee, it is possible to obtain the number of his/her address
- ☐ Given an employee, it is possible to obtain his/her unmarried last name
- ☐ Given an employee, it is possible to obtain his/her last name

Q4 Focusing on classes Project and Company, which of the following entries is true?

- ☐ A company has only one location
- ☐ A project has only one location
- ☐ A company may have several locations

Q5 Focusing on class Employee and its relationships, which of the following entries is true?

- ☐ An employee has several supervisor
- ☐ An employee supervises several employees
- ☐ An employee may have no supervisor

Q6 Focusing on class Company and its relationships, which of the following entries is true?

- ☐ An employee may work for several companies
- ☐ A company is led by only one employee
- ☐ A company may perform no project

Q7 Focusing on classes Resource, Employee, and Project, which of the following entries is true?

- ☐ An employee works for only one project
- ☐ It is possible to obtain the number of hours spent by an employee of a given resource for a given project
- ☐ One or more resources are allocated on a project

Q8 Focusing on class Resource, which of the following entries is true?

- ☐ It is possible to obtain the cost of a software resource
- ☐ A resource may have a hardware component and a software component
- ☐ A resource has a warrantee

Q9 Focusing on classes Employee and Dependent, which of the following entries is true?

- ☐ It is possible to obtain the degree of kinship between an employee and a dependent
- ☐ An employee may have no dependent
- ☐ A dependent is identified by his/her name

Q10 Focusing on the relationship between Employee and Company, which of the following entries is true?

- ☐ An employee may work for no company
- ☐ An employee works for a company and several employees work for a company
- ☐ An employee works for at least a company and at least one employee works for a company

End time:

Final questionnaire

The following questions have to answered after the execution of the comprehension tasks

A1 I had enough time to perform the tasks

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

A2. The task objectives were perfectly clear to me.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

A3. The tasks I performed were perfectly clear to me.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

A4. Judge the difficulty of the comprehension task

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree